

Le système international d'unités The International System of Units

1 édition 2003

Annexe 2

Annex 2

Valeurs recommandées des fréquences étalons
destinées à la mise en pratique de la définition du
mètre et aux représentations secondaires de la
seconde

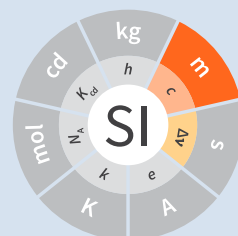
**Recommended values of standard frequencies
for applications including the practical
realization of the metre and secondary
representations of the second**

Partie 2

Part 2

Methane () (2003)

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**Recommended values of standard
frequencies for applications including
the practical realization of the metre and
secondary representations of the second
Part 2: Methane () (2003)**

CCL-CCTF Frequency Standards Working Group

1st edition 2003

1. CIPM recommended values

1.1.

The values $f = 88\,376\,181\,600.18\text{ kHz}$
 $\lambda = 3\,392\,231\,397.327\text{ fm}$

with a relative standard uncertainty of 3×10^{-12} apply to the radiation of a He-Ne laser stabilized to the central component, (7-6) transition, of the resolved hyperfine-structure triplet. The values correspond to the mean frequency of the two recoil-split components for molecules which are effectively stationary, i.e. the values are corrected for second-order Doppler shift.

1.2.

The values $f = 88\,376\,181\,600.5\text{ kHz}$
 $\lambda = 3\,392\,231\,397.31\text{ fm}$

with a relative standard uncertainty of 2.3×10^{-11} apply to the radiation of a He-Ne laser stabilized to the centre of the unresolved hyperfine-structure of a methane cell, within or external to the laser, held at room temperature and subject to the following conditions:

- methane pressure $\leq 3\text{ Pa}$;
- mean one-way intracavity surface power density (i.e., the output power density divided by the transmittance of the output mirror) $\leq 10^4\text{ Wm}^{-2}$;
- radius of wavefront curvature $\geq 1\text{ m}$;
- inequality of power between counter-propagating waves $\leq 5\%$;
- servo-referenced to a detector placed at the output facing the laser tube.

2. Source data

2.1.

Resolved hyperfine structure

Adopted value $f = 88\,376\,181\,600.18\,(27)\,\text{kHz}$

$$u_c/y = 3 \times 10^{-12}$$

for which:

$$\lambda = 3\,392\,231\,397.327\,(10)\,\text{fm}$$

$$u_c/y = 3 \times 10^{-12}$$

calculated from

x/ kHz	Laser	Frequency chain	Year	source data
600.29	LPI	PTB	1991	[1]
599.9	LPI	VNIIFTRI	1985-1986	[2]
600.11	LPI	VNIIFTRI	1989-1992	[2]
600.18	PTB	VNIIFTRI	1989	[2]
600.16	PTB	PTB	1992	[3]
600.44	ILP	ILP	1988-1991	[4]

Unweighted mean $f = 88\,376\,181\,600.18\,\text{kHz}$

where $f = (88\,376\,181\,000 + x)\,\text{kHz}$.

Other available values having uncertainties larger than 200 Hz have not been used. The relative standard uncertainty of one measurement was estimated to be 2.9×10^{-12} using the maximum deviation from the mean and rounded to 3×10^{-12} .

2.2.

Unresolved hyperfine structure

Adopted value $f = 88\,376\,181\,600.5\,(2.0)\,\text{kHz}$

$$u_c/y = 2.3 \times 10^{-11}$$

for which:

$$\lambda = 3\,392\,231\,397.31\,(8)\,\text{fm}$$

$$u_c/y = 2.3 \times 10^{-11}$$

calculated from

x/ kHz	Frequency source	Frequency chain	Year	source data
600.9	Stationary device	ILP	1983	[4], [5], [6], [7]

x/ kHz	Frequency source	Frequency chain	Year	source data
601.48	Portable laser 2	NRC	1985	[8], [9]
599.33	Portable laser 3	NRC	1986-1991	[8], [9]
596.82	Portable laser 1	AIST	1988-1990	[9]
601.52	CH ₄ beam	PTB	1987-1989	[9], [10], [11]
601.77	Portable laser M101	VNIIFTRI	1985-1992	[2], [9]
600.12	Portable laser P1	VNIIFTRI	1985-1988	[2], [9]
598.5	Portable laser PL	VNIIFTRI	1986	[2]
600.96	Portable laser B.3	BIPM	1985-1992	[9]
601.33	Portable laser VB	BIPM	1988-1991	[9]
600.3	Portable laser VNIBI	BIPM	1991	[9], [10], [11], [12]
Unweighted mean:		$f = 88\,376\,181\,600.46$ kHz		
where $f_{\text{CH}_4} = (88\,376\,181\,000 + x)$ kHz.				

The standard deviation of one determination is 1.7 kHz. This is equivalent to a relative uncertainty of 1.9×10^{-11} , increased by the CCL to 2.3×10^{-11} to give an uncertainty of 2 kHz.

Absorbing molecule CH₄, F₂⁽²⁾ component, P(7) ν_3 transition

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